IDSI31135 · ESSEC MiM 1 · FALL 2026

Managing IT in the Digital Age

Complete revision notes · Jean-Noël Tronc

ASSESSMENT

Examen final écrit, papier : 45 % (mardi 8 décembre, 16h30–17h30 d'après les slides de la séance 1 ; vrai/faux, QCM et questions ouvertes ; cheat sheet de 2 pages autorisée). Travaux de groupe en classe : 30 %. Participation orale : 25 %. Présence obligatoire.

CONTENTS

1. S1 · Introduction and digital transformation (6 Oct.)
2. S2 · IS strategy and the Digital Matrix (13 Oct.)

01 S1 · Introduction and digital transformation (6 Oct.)

After class: rewatch video 1 "What is the Digital Transformation and Why Does it Matter" and take the "Digital Transformation" quiz.

What the course aims to do

- Explain what digital transformation means for organisations.
- Understand how it changes the **managing of IS** (strategy, governance, projects, vendors) and **managing through IS** (cloud, enterprise systems, analytics, platforms, AI).
- Make recommendations on transformation strategies, governance, projects, new technologies and their societal and ethical implications.

A brief history of the digital age (slides)

Stage	Key points
Mainframes	Supercomputers, centralised computing
1st computing revolution	The personal computer
Internet	**TCP/IP** protocol (Vinton Cerf, ARPA, 1974); **World Wide Web** developed at CERN by Tim Berners-Lee and Robert Cailliau from 1989, made public in 1993; browsers in the mid-1990s
1st convergence	Telecoms + computing = "telematics" (the French Minitel, 1982–2012)
9 January 2007	Launch of the iPhone: the computer industry takes control of telecoms
Today	Dominance of the **GAFAM** (Google, Apple, Facebook/Meta, Amazon, Microsoft) and the Chinese **BATX** (Baidu, Alibaba, Tencent, Xiaomi)

Quotes to know (slides)

- "Every company is a tech company now" (Satya Nadella, Microsoft).
- "Data is the new oil" (Clive Humby).
- "Mobile will be [a] universal remote control" (Hans Snook, founder of Orange).
- "What's dangerous is not to evolve" (Jeff Bezos).
- "AI will probably most likely lead to the end of the world, but in the meantime, there'll be great companies" (Sam Altman, OpenAI).
- The idea attributed to Darwin: it is not the strongest that survives, but the most adaptable to change.

IT means innovation and robustness: the COBOL example

Course quiz: is COBOL still used? Yes. According to the slides, in 2026 it still accounts for 70 to 80% of the active code in critical business systems (banking, insurance, government), more than 800 billion lines in production, about 95% of card and ATM transactions, and some $3 trillion in daily flows. Lesson: an information system combines new technologies and robust **legacy** that must be managed.

Session objectives (slides)

- Understand the course's teaching philosophy: "Tell me and I will forget, show me and I may remember, involve me and I will understand".
- Know the course rules and session structure.
- Explain what information systems are.
- Identify how technology companies have taken power over the last 25 years ("Tech Companies Take Over").

The syllabus (slides)

Session	Topic	Before / after
1 · 6 Oct.	Introduction and digital transformation	After: video 1, Digital Transformation quiz
2 · 13 Oct.	IS strategy and the Digital Matrix	Before: videos 2 and 3 · after: Digital Matrix quiz
3 · 27 Oct.	IT vendors: sourcing and cloud	Nothing
4 · 3 Nov.	Traditional vs agile IT projects, enterprise systems	Readings
5 · Sat. 7 Nov.	Data analytics and AI	Videos 4 and 5
6 · 10 Nov.	Personal data, cybersecurity, cyber-resilience	Video 6
7 · 17 Nov.	Platform economics	Videos 7, 8, 9 · after: Platforms quiz
8 · 24 Nov.	Cyber-regulation	Readings
9 · 1 Dec.	IT, AI and sustainability	Readings
10 · 8 Dec.	Final exam (paper)	2-page cheat sheet

Course rules

- Attendance is mandatory: beyond 2 absences, even justified, the course is not validated. Be on time (allow for the RER), come back from the break on time.
- Come prepared (videos, readings), participate, take notes on paper; devices only for planned activities (mindfulness, Langer 2000; according to Madore and Wagner 2019, digital multitasking slows you down, tires you and weakens attention and memory).
- In-class group work, fixed groups drawn at random (you are in **group E**); names and group number on the first slide; no mere summaries and no AI-generated content.

02 S2 · IS strategy and the Digital Matrix (13 Oct.)

Before class: videos 2 and 3. After class: "Digital Matrix" quiz.

Session goals

- Explain what information systems are.
- Analyse the digital transformation.
- Explain what strategy and competitive advantage mean in the digital age.
- Apply general tools for strategic analysis.
- Explain and apply the Digital Matrix to understand digital transformation processes.

Key message: digital transformation is first and foremost a **managerial** problem, not a technical one (see the two articles from session 1).

What is an information system?

An IS combines **technology** (software, hardware, techniques including AI, methods, tools, algorithms, programming languages, data processing), **tasks** (manufacturing, marketing, sales, accounting, R&D, project management…) and **people** (employees, managers…).

IT spending stays at a high level (Gartner) while performance keeps growing: **Moore's law**.

Business strategy and IS

A business strategy is "a well-articulated vision of where a business seeks to go and how it expects to get there" (Pearlson and Saunders, 2006). It covers products and services, the industries the company competes in, stakeholders (competitors, suppliers, customers) and long-term goals.

Questions for CEOs: how will digital technologies affect our strategic position? Do "digital" companies threaten us, and which ones? How can incumbents manage the digital transformation successfully?

The digital transformation

- Definition: the integration of digital technology, often called **SMACIT** (Social, Mobile, Analytics, Cloud, Internet of Things), into all areas of a business, resulting in fundamental changes.
- Top objectives of digital strategies (IDG, 2019, 702 respondents): better customer experience (67%), process efficiency through automation (53%), new revenue (48%), keeping pace with competitors (47%).
- Worldwide spending on digital transformation: from USD 0.96 trillion in 2017 to a forecast 2.3 trillion in 2023 (IDC, Statista).
- Faster innovation, globalisation and growing complexity ("Data Never Sleeps"); faster diffusion of innovations.
- Reminder: the transformation affects both **managing of IS** and **managing through IS**.

Competitive advantage

- Barney (1991): an organisation has a competitive advantage when it implements a value-creating strategy not simultaneously implemented by any current or potential competitor.
- Porter (1985): a condition that enables an organisation to operate more efficiently or with higher quality than its competitors.
- Example: Amazon, a tech entrepreneur turned tech giant, innovative on all fronts, from e-commerce to AWS and space.

Analysis tool: Porter's five forces

Industry competitiveness depends on rivalry among existing firms, the bargaining power of suppliers, the bargaining power of buyers, the threat of substitutes and the threat of new entrants (Valacich and Schneider, 2016). IT can affect

each force. Example: in January 2007 Steve Jobs "kills his own market" (the iPod) with the iPhone and disrupts the mobile industry.

America's most valuable companies, 1990 vs 2024: Exxon, Coca-Cola, Walmart… at the top in 1990; Apple (USD 3 trillion), Microsoft, Alphabet, Amazon, NVIDIA, Meta, Tesla today.

The Digital Matrix (Venkatraman)

A grid crossing **three sets of players** and **three phases**, illustrated in the slides with BNP Paribas.

Players	Characteristics	Banking example
Industry incumbents	Your historical competitors: you understand and can anticipate their moves and have a repertoire of responses	HSBC, Bank of America, Citigroup, Crédit Agricole, Société Générale…
Tech entrepreneurs	Ambitious born-digital start-ups that aim to disrupt an industry; algorithms and automation, data as a differential resource, analytics as a differential competency; deep vertical expertise (fintech, autotech…); potential competitors or partners	Fintechs: PayPal, Ant Financial, JD Finance, VaultOS
Digital giants	Yesterday's tech entrepreneurs, now large digital companies; horizontal expertise extended to new domains	Google Wallet, Apple Pay, Libra, Samsung Pay

Phase	What happens	Banking example
1. Experimentation at the edge	Embryonic phase: ideas, prototypes, pilots; scan for digital ideas that could challenge the assumptions of your business model	VaultOS, a cloud-based, blockchain-run OS for banking transactions
2. Collision at the core	Experiments become business options ("could it work?" → "how to make it work profitably?"); digital products and business models challenge traditional ones	PayPal and Bitcoin for international transactions and online shopping
3. Reinvention at the root	Everything is digital; the three sets of players work together on core problems: who interacts with the end customer, who designs the interface, who gathers and analyses the data	BNP and Apple Pay?

The Digital Matrix is **not** a matrix of technologies (it captures players' actions, "business pull", not devices, "business push"), **not** static (actions, reactions, follow-on decisions) and **not** one-dimensional (actions in every phase invite responses from all three sets of players).

The Rules Matrix: three winning moves

1. **Orchestrate and participate**: leverage your core strengths and build value by partnering in an ecosystem, connecting businesses digitally across industries.
2. **Co-creation of capabilities**: collaborate with allies and even competitors to create new offerings and develop capabilities, blurring competition and cooperation.
3. **Amplify human talent with powerful machines**: use machine learning and AI to solve complex problems, guide strategic decisions and reshape work.

Summary (from the slides)

- The five forces are a tool for strategic analysis aimed at competitive advantage; IT can affect each force and IS are often essential to competitive advantage.
- The Digital Matrix helps incumbents succeed in the digital transformation: understand the moves of two distinct types of competitors and respond to the opportunities and threats of digital.